

# IMAT MOCK EXAM 5 — "The Discriminator"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 42/90.0 · Mock exam · answer key + worked solutions at the end

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## Section A - Logical Reasoning & Reading (Q1-9)

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1. "In oxidative phosphorylation, the proton gradient across the inner mitochondrial membrane is coupled to ATP synthesis. If the membrane is made permeable to protons by an uncoupler such as dinitrophenol, the gradient dissipates, electron transport accelerates, but ATP synthesis falls. Heat is produced instead."

Which is the strongest inference?

- A) Uncouplers raise ATP yield.
- B) The proton gradient is independent of electron transport.
- C) Uncoupling disconnects electron transport from ATP synthesis.
- D) Without a proton gradient, ATP synthesis rises.
- E) Uncoupling stops electron transport.

2. "A population bottleneck occurs when a population's size is drastically reduced for at least one generation. The genetic diversity of the surviving population is typically much lower than that of the original, because alleles are lost by chance. After the bottleneck, even rare alleles that remain can rise in frequency by genetic drift."

Which follows?

- A) Genetic drift stops after a bottleneck.
- B) Bottlenecks have no evolutionary consequence.
- C) Bottlenecks reduce genetic diversity regardless of which alleles happen to survive.
- D) After a bottleneck, the surviving diversity depends partly on which alleles randomly survived.
- E) Bottlenecks always raise genetic diversity.

3. (General knowledge) The Italian physician and educator who developed an eponymous educational method emphasising self-directed activity in a prepared environment, and was nominated for the Nobel Peace Prize multiple times, was:

- A) Rita Levi-Montalcini
- B) Maria Montessori
- C) Elena Lucrezia Cornaro Piscopia
- D) Sofia Vanni
- E) Nilde Iotti

4. Five people — F, G, H, I, J — are in a queue. F is immediately behind G. H is somewhere in front of G. I is not at either end. J is at the back. Which must be true?

- A) G is at the front.
- B) I is between H and G.
- C) F is immediately in front of J.
- D) F is third.
- E) H is at the front.

5. "All metals conduct electricity. X conducts electricity." From these, we can conclude:

- A) X might or might not be a metal (affirming the consequent is invalid).
- B) X is a metal.
- C) No metal is X.
- D) X is not a metal.
- E) X is an insulator.

6. A number, when divided by 3, gives remainder 2; when divided by 5, gives remainder 3. The smallest positive such number is:

- A) 5   B) 7   C) 8   D) 11   E) 13

7. A clock reads 8:20. What is the angle between the hands?  
**A)** 0° **B)** 100° **C)** 120° **D)** 130° **E)** 140°
8. If the 5th of a month is a Tuesday, what day is the 27th of the same month?  
**A)** Tuesday **B)** Wednesday **C)** Thursday **D)** Saturday **E)** Friday
9. Complete the letter series: A, C, F, J, O, ?  
**A)** S **B)** T **C)** V **D)** W **E)** U

## **Section B - Biology (Q10-32)**

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10. In glycolysis, the committed (rate-limiting) step is catalysed by:  
**A)** Phosphofructokinase-1 (PFK-1)  
**B)** Aldolase  
**C)** Phosphoglycerate kinase  
**D)** Hexokinase  
**E)** Pyruvate kinase
11. In the electron transport chain, FADH<sub>2</sub> donates electrons to:  
**A)** Complex II (succinate dehydrogenase)  
**B)** Complex IV  
**C)** ATP synthase  
**D)** Complex III  
**E)** Complex I
12. The chemiosmotic synthesis of ATP in mitochondria is driven by the flow of protons through ATP synthase from:  
**A)** Cytosol to matrix  
**B)** Intermembrane space to matrix  
**C)** Matrix to intermembrane space  
**D)** Outer to inner membrane (no gradient)  
**E)** Matrix to cytosol
13. Under anaerobic conditions, yeast cells regenerate NAD<sup>+</sup> by reducing pyruvate to:  
**A)** Water  
**B)** Acetyl-CoA  
**C)** Ethanol and CO<sub>2</sub>  
**D)** Glucose  
**E)** Lactate
14. During muscle contraction, calcium ions released from the sarcoplasmic reticulum bind to:  
**A)** Myosin heads  
**B)** Actin directly  
**C)** The sarcolemma only  
**D)** ATP  
**E)** Troponin, causing tropomyosin to move and expose myosin-binding sites
15. The sliding filament theory states that during contraction:  
**A)** The Z-lines move apart.  
**B)** Actin filaments shorten.  
**C)** Myosin filaments shorten.  
**D)** The sarcomere shortens, but the filaments themselves do not change length (they slide past each other).  
**E)** Both filaments shorten in proportion.

- 16.** Slow-twitch (Type I) muscle fibres are specialised for:
- A)** Replacing damaged neurons
  - B)** Generating action potentials
  - C)** Storing calcium
  - D)** Rapid, powerful, short-duration contractions using anaerobic glycolysis
  - E)** Endurance activity using oxidative metabolism (many mitochondria, high myoglobin)
- 17.** Most CO<sub>2</sub> transported in the blood is carried:
- A)** Dissolved as CO<sub>2</sub> gas in plasma
  - B)** As bicarbonate ions (HCO<sub>3</sub><sup>-</sup>) in plasma
  - C)** Bound to albumin
  - D)** Within erythrocyte nuclei
  - E)** Bound to haemoglobin (carbaminohaemoglobin)
- 18.** The enzyme carbonic anhydrase, abundant in red blood cells, catalyses:
- A)**  $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$
  - B)** Glycolysis
  - C)** The conversion of haemoglobin to oxyhaemoglobin
  - D)** Pyruvate → acetyl-CoA
  - E)** The clotting cascade
- 19.** In the nephron, the majority of filtered water and solutes is reabsorbed in the:
- A)** Loop of Henle (descending limb only)
  - B)** Distal convoluted tubule
  - C)** Collecting duct
  - D)** Proximal convoluted tubule (~65%)
  - E)** Renal pelvis
- 20.** Antidiuretic hormone (ADH) increases water reabsorption by inserting aquaporins into the:
- A)** Afferent arteriole
  - B)** Proximal tubule
  - C)** Collecting duct (principal cells)
  - D)** Glomerulus
  - E)** Loop of Henle (thin limb)
- 21.** Aldosterone, secreted by the adrenal cortex, acts on the kidney to increase reabsorption of:
- A)** Glucose
  - B)** Urea
  - C)** K<sup>+</sup>
  - D)** Albumin
  - E)** Na<sup>+</sup> (and water), and increase K<sup>+</sup>/H<sup>+</sup> excretion
- 22.** In nervous tissue, the myelin sheath is produced in the CNS by \_\_ and in the PNS by \_\_.
- A)** Oligodendrocytes / Schwann cells
  - B)** Ependymal cells / astrocytes
  - C)** Astrocytes / Schwann cells
  - D)** Microglia / satellite cells
  - E)** Neurons / oligodendrocytes
- 23.** During an action potential, the rapid depolarisation is due to:
- A)** K<sup>+</sup> efflux
  - B)** Ca<sup>2+</sup> influx at the soma
  - C)** Na<sup>+</sup> influx (voltage-gated Na<sup>+</sup> channels open)
  - D)** Cl<sup>-</sup> influx
  - E)** Active transport of Na<sup>+</sup> out

- 24.** A cross between two plants both with red flowers produces 3 red : 1 white offspring. This is consistent with:
- A)** Polygenic inheritance
  - B)** Complete dominance (red dominant, parents heterozygous)
  - C)** Sex-linkage
  - D)** Incomplete dominance
  - E)** Codominance
- 25.** A man with blood group A (genotype AO) and a woman with blood group B (genotype BO) can have children with which blood groups?
- A)** O only
  - B)** A only
  - C)** B only
  - D)** A, B, AB, and O
  - E)** AB only
- 26.** DNA polymerase synthesises new DNA in which direction?
- A)** 3' → 5' on the new strand
  - B)** From the C-terminus
  - C)** Without direction
  - D)** In both directions simultaneously on one strand
  - E)** 5' → 3' on the new strand
- 27.** The genetic code is described as "degenerate" because:
- A)** Codons overlap.
  - B)** One codon codes for multiple amino acids.
  - C)** The code is ambiguous (one codon → multiple amino acids).
  - D)** Multiple codons can code for the same amino acid.
  - E)** The code differs between species.
- 28.** A bacterial operon whose transcription is increased in the presence of a small molecule (e.g., lactose for the lac operon) is:
- A)** Inducible (the molecule is an inducer)
  - B)** Constitutive
  - C)** Repressible
  - D)** Always off
  - E)** Negatively regulated only
- 29.** Which cell type produces antibodies?
- A)** Macrophages
  - B)** Platelets
  - C)** B cells (plasma cells, the differentiated form)
  - D)** Mast cells
  - E)** T-helper cells
- 30.** In photosynthesis, the oxygen released comes from:
- A)** O<sub>2</sub> taken from the air
  - B)** H<sub>2</sub>O (water, split in the light reactions)
  - C)** Glucose
  - D)** CO<sub>2</sub>
  - E)** NADPH
- 31.** In a DNA double helix, 20% of bases are adenine. The percentage of guanine is:
- A)** 20%   **B)** 30%   **C)** 40%   **D)** 60%   **E)** 80%

32. A mutation that inserts a single nucleotide into a coding sequence (not in multiples of 3) is called a:
- A) Point mutation only
  - B) Nonsense mutation
  - C) Silent mutation
  - D) Missense mutation
  - E) Frameshift mutation

### Section C - Chemistry (Q33-47)

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33. How many grams of  $\text{CO}_2$  are produced when 0.5 mol of C reacts completely with  $\text{O}_2$ ? (C = 12, O = 16.)
- A) 4.4 g   B) 11 g   C) 22 g   D) 44 g   E) 88 g
34. Balance:  $\_\_\text{Fe} + \_\_\text{O}_2 \rightarrow \_\_\text{Fe}_2\text{O}_3$ . The coefficient of Fe is:
- A) 1   B) 2   C) 3   D) 4   E) 6
35. What volume of 0.1 M  $\text{H}_2\text{SO}_4$  is needed to neutralise 25 mL of 0.2 M NaOH? ( $\text{H}_2\text{SO}_4$  is diprotic.)
- A) 5 mL   B) 12.5 mL   C) 25 mL   D) 50 mL   E) 100 mL
36. The pH of a buffer containing 0.1 M acetic acid ( $\text{pK}_a = 4.76$ ) and 0.1 M sodium acetate is:
- A) 3.76   B) 4.76   C) 5.76   D) 7.0   E) 9.24
37. The oxidation number of oxygen in hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) is:
- A) -2   B) -1   C) 0   D) +1   E) +2
38. Which species is reduced in:  $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ ?
- A)  $\text{Cl}_2$  ( $\text{Cl}_2 \rightarrow 2\text{Cl}^-$ )
  - B) Na ( $\text{Na} \rightarrow \text{Na}^+$ )
  - C) Nothing
  - D) NaCl
  - E) Both Na and  $\text{Cl}_2$
39. An ideal gas occupies 2 L at 1 atm and 300 K. What is its volume at 2 atm and 300 K?
- A) 0.5 L   B) 1 L   C) 2 L   D) 4 L   E) 8 L
40. How many electrons can occupy the d-subshell (5 orbitals)?
- A) 2   B) 6   C) 8   D) 10   E) 14
41. Which element has the electron configuration  $1s^2 2s^2 2p^6 3s^1$ ?
- A) Na   B) K   C) Mg   D) Al   E) Ne
42. The geometry of  $\text{NH}_3$  (ammonia) is:
- A) Trigonal planar
  - B) Octahedral
  - C) Square planar
  - D) Linear
  - E) Tetrahedral (electron-pair geometry) / trigonal pyramidal (molecular shape)
43. Which compound is most soluble in water?
- A)  $\text{CH}_3\text{OH}$  (methanol)
  - B)  $\text{CCl}_4$
  - C)  $\text{CH}_4$  (methane)
  - D)  $\text{C}_6\text{H}_6$  (benzene)
  - E)  $\text{C}_2\text{H}_6$  (ethane)

44. An exothermic reaction has:  
**A)** No activation energy  
**B)** Products higher in energy than reactants  
**C)**  $\Delta H > 0$   
**D)**  $\Delta H = 0$   
**E)**  $\Delta H < 0$
45. The IUPAC name of  $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$  is:  
**A)** Propane **B)** Ethane **C)** Methane **D)** Butane **E)** Pentane
46. For the equilibrium  $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ ,  $\Delta H < 0$ . Increasing pressure at constant T will:  
**A)** Shift left  
**B)** Shift right  
**C)** Have no effect (equal moles of gas on both sides)  
**D)** Decrease the rate  
**E)** Increase  $\Delta H$
47. If the rate constant doubles when temperature increases by  $10^\circ\text{C}$ , and the rate at  $T_1$  is  $r$ , the rate at  $T_1 + 20^\circ\text{C}$  is approximately:  
**A)**  $2r$  **B)**  $3r$  **C)**  $4r$  **D)**  $8r$  **E)**  $10r$

### Section D - Mathematics (Q48-54)

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48. Solve:  $x^2 - 5x - 6 = 0$ .  
**A)** No real roots **B)**  $x = 1, -6$  **C)**  $x = 2, 3$  **D)**  $x = 5, 1$  **E)**  $x = -1, 6$
49. Solve:  $\log_3(x) = -2$ .  
**A)**  $x = 1/9$  **B)**  $x = 9$  **C)**  $x = -2$  **D)**  $x = -6$  **E)**  $x = 1/3$
50. Solve:  $(2x - 1)/(x + 1) = 3$  ( $x \neq -1$ ).  
**A)**  $x = 1$  **B)**  $x = -2$  **C)**  $x = 2$  **D)**  $x = -4$  **E)**  $x = 4$
51. Solve the inequality:  $x^2 - 5x + 6 > 0$ .  
**A)**  $2 < x < 3$  **B)**  $x > 2$  **C)**  $x < 2$  or  $x > 3$  **D)**  $x < 3$  **E)** All real  $x$
52. A circle has circumference  $12\pi$ . What is its area?  
**A)**  $6\pi$  **B)**  $12\pi$  **C)**  $24\pi$  **D)**  $36\pi$  **E)**  $144\pi$
53. In a right triangle, the side opposite the  $30^\circ$  angle has length 5. What is the length of the hypotenuse?  
**A)** 5 **B)**  $5\sqrt{2}$  **C)**  $5\sqrt{3}$  **D)** 10 **E)** 15
54. Two dice are rolled. What is the probability that the sum is 7?  
**A)**  $1/12$  **B)**  $1/9$  **C)**  $1/6$  **D)**  $5/36$  **E)**  $7/36$

### Section E - Physics (Q55-60)

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55. A car starts from rest and accelerates uniformly at  $3 \text{ m/s}^2$ . How far does it travel in 4 s?  
**A)** 12 m **B)** 24 m **C)** 36 m **D)** 48 m **E)** 72 m
56. A  $6 \Omega$  and a  $3 \Omega$  resistor are connected in parallel across a 12 V battery. What is the total current from the battery?  
**A)** 1 A **B)** 2 A **C)** 3 A **D)** 4 A **E)** 6 A
57. A 100 W bulb is operated on a 200 V supply for 10 s. How much electrical energy is used?  
**A)** 20,000 J **B)** 100 J **C)** 200 J **D)** 1000 J **E)** 2000 J

**58.** A 2 kg ball moving at 5 m/s strikes a wall and rebounds at 3 m/s. What is the magnitude of the change in momentum?

- A)** 2 kg·m/s   **B)** 4 kg·m/s   **C)** 8 kg·m/s   **D)** 10 kg·m/s   **E)** 16 kg·m/s

**59.** A satellite orbits Earth at a radius where gravitational acceleration is  $2 \text{ m/s}^2$ . If Earth's radius is  $R$  and the satellite is at  $3R$  from the centre, the surface gravitational acceleration is:

- A)**  $2 \text{ m/s}^2$    **B)**  $6 \text{ m/s}^2$    **C)**  $9 \text{ m/s}^2$    **D)**  $18 \text{ m/s}^2$    **E)**  $27 \text{ m/s}^2$

**60.** A 0.5 kg block of iron ( $c = 450 \text{ J/kg} \cdot ^\circ\text{C}$ ) is heated from  $20^\circ\text{C}$  to  $100^\circ\text{C}$ . How much heat is required?

- A)** 9,000 J   **B)** 18,000 J   **C)** 22,500 J   **D)** 45,000 J   **E)** 90,000 J

## Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

|    |   |    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|----|---|
| 1  | C | 2  | D | 3  | B | 4  | E | 5  | A |
| 6  | C | 7  | D | 8  | B | 9  | E | 10 | A |
| 11 | A | 12 | B | 13 | C | 14 | E | 15 | D |
| 16 | E | 17 | B | 18 | A | 19 | D | 20 | C |
| 21 | E | 22 | A | 23 | C | 24 | B | 25 | D |
| 26 | E | 27 | D | 28 | A | 29 | C | 30 | B |
| 31 | B | 32 | E | 33 | C | 34 | D | 35 | C |
| 36 | B | 37 | B | 38 | A | 39 | B | 40 | D |
| 41 | A | 42 | E | 43 | A | 44 | E | 45 | D |
| 46 | C | 47 | C | 48 | E | 49 | A | 50 | D |
| 51 | C | 52 | D | 53 | D | 54 | C | 55 | B |
| 56 | E | 57 | D | 58 | E | 59 | D | 60 | B |

## Worked solutions

Each solution shows the reasoning that produces the key, and names the wrong options by their content. Letters refer to the option order printed in this paper.

### Question 1 • Answer: C

The passage lists three simultaneous effects of the uncoupler: the proton gradient dissipates, electron transport accelerates, and ATP synthesis falls as the energy escapes as heat. The only statement consistent with all three is that uncoupling breaks the coupling between electron transport and ATP synthesis: oxidation runs on, even faster, while phosphorylation stops. The claim that uncoupling stops electron transport is directly contradicted by the stated acceleration, while the claims that ATP synthesis rises without the gradient, that uncouplers raise ATP yield, and that the gradient is independent of electron transport all contradict the fall in ATP output.

### Question 2 • Answer: D

The passage states that alleles are lost by chance during the bottleneck and that surviving rare alleles may afterwards rise by drift, so the diversity that persists and evolves depends partly on the random subset of alleles that survived. The claim that bottlenecks always raise diversity contradicts the stated loss of alleles, and saying drift stops afterwards contradicts the passage's explicit role for drift. Even the broader claim that diversity falls regardless of which alleles survive misses the point: the outcome hinges on which alleles chance kept, which is exactly what the correct inference says.

### Question 3 • Answer: B

Maria Montessori (1870 to 1952) developed the Montessori method, built on self-directed activity within a prepared environment, and was nominated several times for the Nobel Peace Prize. Rita Levi-Montalcini was a Nobel laureate neuroembryologist and Elena Lucrezia Cornaro Piscopia the first woman to receive a university doctorate, so neither matches the educational-method description.

### Question 4 • Answer: E

Number the positions 1 (front) to 5 (back). J sits at 5, and since F is immediately behind G the pair G,F occupies either positions 3,4 or 2,3. Step 1: if G is third and F fourth, places 1 and 2 go to H and I; I cannot take place 1 (an end), so I takes place 2 and H is forced to the front (order H, I, G, F, J). Step 2: if G is second and F third, H must be somewhere in front of G, so H again takes place 1 and I takes place 4, which is not an end (order H, G, F, I, J). In both admissible orders H is first, so the statement that must be true is that H is at the front. Note that F immediately in front of J fails in the second order, since I rather than F precedes J there.

### Question 5 • Answer: A

The premise gives only metal  $\rightarrow$  conducts electricity. Being told that X conducts affirms the consequent, which is an invalid inference: graphite and electrolyte solutions conduct without being metals. So X might or might not be a metal; the claims that X is a metal, not a metal, or an insulator all overstate what follows (an insulator would even contradict the given fact).

### Question 6 • Answer: C

Numbers leaving remainder 3 on division by 5 are 3, 8, 13, 18, ... Testing in order: 3 leaves remainder 0 on division by 3, failing;  $8 = 3 \times 2 + 2$  leaves remainder 2 on division by 3 and  $8 = 5 \times 1 + 3$  leaves remainder 3 on division by 5. Both conditions are met, so 8 is the smallest positive solution. Among the offered values none of the others passes both tests (5 leaves remainder 0 on division by 5, 7 and 13 leave remainder 1 on division by 3, 11 leaves remainder 1 on division by 5), so the solution is unique.

### Question 7 • Answer: D

The minute hand moves  $6^\circ$  per minute, so at 20 minutes past it sits at  $20 \times 6 = 120^\circ$  measured from 12. The hour hand moves  $30^\circ$  per hour plus  $0.5^\circ$  per minute, so it sits at  $8 \times 30 + 20 \times 0.5 = 240 + 10 = 250^\circ$ . The angle between the hands is  $250 - 120 = 130^\circ$ . The tempting  $120^\circ$  answer forgets that the hour hand has crept  $10^\circ$  past the 8 during those 20 minutes.

**Question 8 • Answer: B**

The 27th is  $27 - 5 = 22$  days after the 5th, and  $22 = 3 \times 7 + 1$ , so the weekday advances by exactly one day: Tuesday plus one day is Wednesday. Reading 22 days as three whole weeks with no leftover day would wrongly return Tuesday.

**Question 9 • Answer: E**

In alphabet positions the terms are A = 1, C = 3, F = 6, J = 10, O = 15. The gaps are +2, +3, +4, +5, so the next gap is +6, giving  $15 + 6 = 21$ , the 21st letter, U. Any constant-step reading (which would give S or T) fits none of the consecutive pairs in the series.

**Question 10 • Answer: A**

The committed, rate-limiting step of glycolysis is the phosphorylation of fructose-6-phosphate to fructose-1,6-bisphosphate by phosphofructokinase-1, the pathway's principal allosteric control point (inhibited by ATP, activated by AMP). Hexokinase acts earlier and pyruvate kinase last, but neither is the committed control point: once PFK-1 has acted, the sugar cannot easily leave the glycolytic route.

**Question 11 • Answer: A**

FADH<sub>2</sub> is reoxidised by succinate dehydrogenase, which is itself Complex II of the chain, so its electrons enter downstream of Complex I and fewer protons are pumped (about 1.5 ATP per FADH<sub>2</sub> versus about 2.5 per NADH). Assigning FADH<sub>2</sub> to Complex I describes NADH, and ATP synthase is not an electron carrier at all.

**Question 12 • Answer: B**

The respiratory chain pumps protons out of the matrix into the intermembrane space, storing energy as an electrochemical gradient. ATP synthase then allows protons to flow back down that gradient, from intermembrane space to matrix, and this return flow powers ATP synthesis. The reverse direction (matrix to intermembrane space) is the pumping step, not the flow through ATP synthase.

**Question 13 • Answer: C**

Yeast performs alcoholic fermentation: pyruvate is decarboxylated to acetaldehyde (releasing CO<sub>2</sub>), and acetaldehyde is reduced to ethanol, oxidising NADH back to NAD<sup>+</sup> so glycolysis can continue. Lactate is the product in oxygen-starved mammalian muscle and in lactic acid bacteria, not in yeast.

**Question 14 • Answer: E**

Ca<sup>2+</sup> released from the sarcoplasmic reticulum binds troponin C; the troponin complex then shifts tropomyosin away from actin's myosin-binding sites, permitting cross-bridge formation. Calcium does not trigger contraction by binding the myosin heads or actin directly.

**Question 15 • Answer: D**

In the sliding filament model the thick and thin filaments keep constant length and simply slide past one another, pulling the Z-lines toward the sarcomere centre; the sarcomere shortens while neither filament does. Each claim that actin or myosin (or both) shortens contradicts the model's central point, and the Z-lines moving apart is the opposite of what happens.

**Question 16 • Answer: E**

Type I (slow-twitch, red) fibres are specialised for endurance: abundant mitochondria, high myoglobin, dense capillaries and predominantly oxidative metabolism make them fatigue-resistant. The description of rapid, powerful, short-duration anaerobic bursts matches fast-twitch (Type II) fibres instead.

**Question 17 • Answer: B**

About 70% of blood CO<sub>2</sub> travels as bicarbonate: CO<sub>2</sub> enters red blood cells, carbonic anhydrase converts it to carbonic acid, which dissociates to H<sup>+</sup> and HCO<sub>3</sub><sup>-</sup>, and the bicarbonate is exported into the plasma. Only about 23% rides on haemoglobin as carbaminohaemoglobin and about 7% dissolves in plasma, so neither of those is the majority route.

**Question 18 • Answer: A**

Carbonic anhydrase catalyses the reversible hydration  $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$ , which then dissociates into  $\text{H}^+$  and  $\text{HCO}_3^-$ ; this reaction underpins  $\text{CO}_2$  transport and the red cell's buffering of blood pH. It has no role in haemoglobin oxygenation, glycolysis, clotting or the pyruvate to acetyl-CoA step.

**Question 19 • Answer: D**

The proximal convoluted tubule reabsorbs roughly 65% of the filtered  $\text{Na}^+$ , water, glucose and amino acids, by far the largest share of any nephron segment. The distal tubule and collecting duct only fine-tune what remains, and the renal pelvis merely drains urine.

**Question 20 • Answer: C**

ADH binds V2 receptors on collecting duct principal cells and triggers insertion of aquaporin-2 channels into the apical membrane, sharply increasing the duct's water permeability and hence reabsorption. The glomerulus filters rather than reabsorbs, and proximal water reabsorption follows solutes osmotically without ADH control.

**Question 21 • Answer: E**

Aldosterone from the adrenal cortex acts on the distal tubule and collecting duct to increase  $\text{Na}^+$  reabsorption (water follows passively) while promoting  $\text{K}^+$  and  $\text{H}^+$  secretion. The option asserting increased  $\text{K}^+$  reabsorption states the opposite of the hormone's actual effect on potassium.

**Question 22 • Answer: A**

In the CNS, oligodendrocytes wrap myelin around axon segments (one cell serving several axons); in the PNS, Schwann cells myelinate single segments. The pairing astrocytes with Schwann cells fails because astrocytes are supporting and blood-brain barrier cells, not myelin formers, and microglia are the CNS's phagocytes.

**Question 23 • Answer: C**

The rapid upstroke of the action potential is produced by the opening of voltage-gated  $\text{Na}^+$  channels:  $\text{Na}^+$  floods in down its electrochemical gradient and drives the potential toward about +30 mV.  $\text{K}^+$  efflux through delayed channels causes the repolarising falling phase, and active  $\text{Na}^+$  extrusion by the  $\text{Na}^+/\text{K}^+$  pump is thousands of times too slow to account for a millisecond-scale depolarisation.

**Question 24 • Answer: B**

A 3 red : 1 white ratio from two red parents is the classic monohybrid result of two heterozygotes under complete dominance: the 1:2:1 genotypic ratio collapses to 3 dominant phenotypes per 1 recessive. Incomplete dominance would instead yield a 1:2:1 phenotype ratio with an intermediate class, and codominance would express both traits side by side; neither produces 3:1.

**Question 25 • Answer: D**

The father (AO) passes on A or O and the mother (BO) passes on B or O, so the offspring genotypes are AB, AO (group A), BO (group B) and OO (group O). All four ABO phenotypes are therefore possible. Any answer restricting the children to a single group discards three of the four equally likely combinations.

**Question 26 • Answer: E**

DNA polymerase can only attach a nucleotide to a free 3'-OH end, so every new strand grows in the 5' to 3' direction while its template is read 3' to 5'. Growth in the 3' to 5' direction on the new strand is chemically impossible, which is exactly why the lagging strand is built discontinuously as Okazaki fragments.

**Question 27 • Answer: D**

The code is degenerate because most amino acids are specified by more than one codon (leucine by six), a consequence of 61 sense codons coding for about 20 amino acids. Degeneracy is not ambiguity: each individual codon still specifies exactly one amino acid, and the code is essentially universal across species.

**Question 28 • Answer: A**

An operon whose transcription increases when a small molecule is present is inducible, and that molecule is the inducer; for the lac operon, allolactose inactivates the lac repressor and transcription rises. A repressible operon behaves oppositely (transcription falls when the co-repressor, e.g. tryptophan for the trp operon, is present), and constitutive means expressed regardless of the molecule.

**Question 29 • Answer: C**

Antibodies are secreted by plasma cells, the terminally differentiated form of B lymphocytes. T-helper cells coordinate the response via cytokines, macrophages phagocytose, and mast cells release histamine, but none of them secretes antibody.

**Question 30 • Answer: B**

The  $O_2$  released comes from splitting water at photosystem II's oxygen-evolving complex:  $2 H_2O \rightarrow 4 H^+ + 4 e^- + O_2$ . Isotope labelling settled it: water labelled with  $^{18}O$  yields labelled  $O_2$ , whereas labelled  $CO_2$  does not, ruling out carbon dioxide or glucose as the oxygen source.

**Question 31 • Answer: B**

By Chargaff's rules  $A = T$  and  $G = C$  in double-stranded DNA. With  $A = 20\%$ ,  $T$  is also  $20\%$ , leaving  $100 - 40 = 60\%$  to split equally between  $G$  and  $C$ , so  $G = 30\%$ . The answer of  $20\%$  would wrongly make all four bases equal, and  $60\%$  is the combined  $G + C$  fraction, not  $G$  alone.

**Question 32 • Answer: E**

Inserting one nucleotide, since 1 is not a multiple of 3, shifts the triplet reading frame for every codon downstream, so this is a frameshift mutation. Missense and nonsense mutations change one codon at a time without disturbing the frame, and silent mutations change no amino acid at all.

**Question 33 • Answer: C**

From  $C + O_2 \rightarrow CO_2$  the mole ratio of carbon to carbon dioxide is 1:1, so 0.5 mol of  $C$  yields 0.5 mol of  $CO_2$ . The molar mass of  $CO_2$  is  $12 + 2 \times 16 = 44$  g/mol, giving  $0.5 \times 44 = 22$  g. The value 11 g would correspond to only 0.25 mol, i.e. misapplying the stoichiometry.

**Question 34 • Answer: D**

Balance by atom counting: each  $Fe_2O_3$  contains 2 Fe and 3 O, so taking 2  $Fe_2O_3$  requires 4 Fe and 6 O atoms, i.e. 3  $O_2$ . The balanced equation is  $4 Fe + 3 O_2 \rightarrow 2 Fe_2O_3$ , so the coefficient of Fe is 4. Coefficients of 2 or 3 come from balancing only one element and forgetting the other must match too.

**Question 35 • Answer: C**

Moles of  $NaOH = 0.025 \text{ L} \times 0.2 \text{ mol/L} = 0.005 \text{ mol}$ , which needs 0.005 mol of  $H^+$  to neutralise. Since each  $H_2SO_4$  donates 2  $H^+$ , the acid required is  $0.005 \div 2 = 0.0025 \text{ mol}$ , so  $V = 0.0025 \div 0.1 = 0.025 \text{ L} = 25 \text{ mL}$ . The value 12.5 mL is the classic slip of treating diprotic sulfuric acid as monoprotic.

**Question 36 • Answer: B**

Henderson-Hasselbalch:  $pH = pK_a + \log([A^-]/[HA]) = 4.76 + \log(0.1/0.1) = 4.76 + 0 = 4.76$ . Whenever a buffer's conjugate base and acid concentrations are equal, its pH equals the  $pK_a$ . The values 3.76 or 5.76 would demand a tenfold ratio between the two components.

**Question 37 • Answer: B**

Hydrogen peroxide is a peroxide, and in peroxides oxygen takes the exceptional oxidation state  $-1$  rather than the usual  $-2$ . Check: with H at  $+1$ ,  $2(+1) + 2x = 0$  gives  $x = -1$ . The value  $-2$  applies to ordinary oxides, and 0 describes elemental oxygen.

**Question 38 • Answer: A**

Reduction is gain of electrons: chlorine falls from oxidation number 0 in  $Cl_2$  to  $-1$  in  $NaCl$ , so  $Cl_2$  is reduced, while sodium rises from 0 to  $+1$  and is oxidised. Claiming both species are reduced ignores that the electrons lost by sodium must be gained by chlorine, and the product  $NaCl$  is not a reacting species.

**Question 39 • Answer: B**

At constant temperature and amount, Boyle's law applies:  $P_1V_1 = P_2V_2$ , so  $1 \text{ atm} \times 2 \text{ L} = 2 \text{ atm} \times V_2$ , giving  $V_2 = 1 \text{ L}$ . Doubling the pressure halves the volume; the answer of 4 L wrongly treats pressure and volume as directly proportional.

**Question 40 • Answer: D**

Each of the five d orbitals holds at most 2 electrons of opposite spin, so the d subshell holds  $5 \times 2 = 10$  electrons. The capacities 6 and 14 belong to the p and f subshells (3 and 7 orbitals respectively).

**Question 41 • Answer: A**

Summing the superscripts:  $2 + 2 + 6 + 1 = 11$  electrons, so the element has atomic number 11, sodium. Its lone  $3s^1$  valence electron explains sodium's +1 chemistry; magnesium would be  $3s^2$  and neon stops at  $2p^6$ .

**Question 42 • Answer: E**

Nitrogen in  $\text{NH}_3$  has three bonding pairs plus one lone pair, four electron domains in all, so the electron-pair geometry is tetrahedral; the lone pair occupies one corner, giving the molecular shape of a trigonal pyramid with a bond angle near  $107^\circ$ . Trigonal planar would require the lone pair's absence, and linear or octahedral arrangements fit neither the domain count nor the observed shape.

**Question 43 • Answer: A**

Methanol carries a polar O-H group and can donate and accept hydrogen bonds with water, making it miscible with water in all proportions. Methane, ethane, benzene and carbon tetrachloride are essentially non-polar and cannot hydrogen-bond with water, so all dissolve far less.

**Question 44 • Answer: E**

An exothermic reaction releases heat to the surroundings, so the products end up lower in enthalpy than the reactants and  $\Delta H$  is negative.  $\Delta H > 0$  is the definition of an endothermic reaction, and every chemical reaction still faces an activation energy barrier regardless of the sign of  $\Delta H$ .

**Question 45 • Answer: D**

The chain  $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$  contains four carbon atoms joined only by single bonds, so the alkane is butane,  $\text{C}_4\text{H}_{10}$ . Propane has three carbons and pentane five.

**Question 46 • Answer: C**

Le Chatelier's response to pressure only occurs when the change alters the total moles of gas. Both sides of  $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$  hold exactly 2 moles of gas, so raising pressure at constant temperature shifts the position of equilibrium not at all (though the rate does increase). Predicting a right or left shift misapplies the principle to a balanced-mole system, and  $\Delta H$  is a fixed property of the reaction, not of the pressure.

**Question 47 • Answer: C**

A rate constant doubling per  $10^\circ\text{C}$  compounds over consecutive  $10^\circ\text{C}$  steps. A  $20^\circ\text{C}$  rise is two such steps, so the rate becomes  $2 \times 2 = 4$  times  $r$ . Taking just one doubling ( $2r$ ) treats the whole  $20^\circ\text{C}$  as a single step.

**Question 48 • Answer: E**

Factor by finding two numbers with product  $-6$  and sum  $-5$ : they are  $-6$  and  $+1$ , so  $x^2 - 5x - 6 = (x - 6)(x + 1) = 0$ , giving  $x = 6$  or  $x = -1$ . The pair  $+1$  and  $-6$  gives the right product but the wrong sign of the sum, the standard factorisation slip here.

**Question 49 • Answer: A**

Convert to exponential form:  $\log_3 x = -2$  means  $x = 3^{-2} = 1/9$ . The argument of a logarithm must be positive, so the negative values offered cannot be solutions, and  $x = 9$  would correspond to  $\log_3 9 = 2$ , not  $-2$ .

**Question 50 • Answer: D**

Multiply both sides by  $(x + 1)$ :  $2x - 1 = 3(x + 1) = 3x + 3$ , so  $2x - 3x = 3 + 1$ , giving  $-x = 4$  and  $x = -4$ . Check:  $(2 \times (-4) - 1)/((-4) + 1) = (-9)/(-3) = 3$ , as required. The value  $+4$  results from a sign slip when collecting the  $x$  terms.

**Question 51 • Answer: C**

Factor:  $x^2 - 5x + 6 = (x - 2)(x - 3)$ . The parabola opens upward and equals zero at  $x = 2$  and  $x = 3$ , so it is positive outside the roots:  $x < 2$  or  $x > 3$ . The interval  $2 < x < 3$  is precisely where the expression is negative, the opposite of what is required.

**Question 52 • Answer: D**

Circumference  $C = 2\pi r = 12\pi$  gives  $r = 6$ , so area  $= \pi r^2 = \pi \times 36 = 36\pi$ . Answering  $12\pi$  reuses the circumference figure as the area, and  $144\pi$  results from squaring 12 (the diameter) instead of 6.

**Question 53 • Answer: D**

In a 30-60-90 triangle the side opposite the  $30^\circ$  angle is half the hypotenuse, since  $\sin 30^\circ = \text{opposite/hypotenuse} = 1/2$ . With that side equal to 5, the hypotenuse is  $2 \times 5 = 10$ . The lengths  $5\sqrt{2}$  and  $5\sqrt{3}$  belong to 45-45-90 triangles and to the longer leg of a 30-60-90 triangle respectively, not to the hypotenuse.

**Question 54 • Answer: C**

Two dice produce 36 equally likely ordered outcomes. A sum of 7 arises from six of them: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), more than any other total. The probability is  $6/36 = 1/6$ ; the value  $5/36$  belongs to totals such as 6 or 8, which have one combination fewer.

**Question 55 • Answer: B**

Starting from rest,  $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 3 \times 4^2 = \frac{1}{2} \times 3 \times 16 = 24$  m. The value 12 m comes from using  $s = at$  and dropping the half, and 48 m from using  $at^2$  without the  $\frac{1}{2}$  factor.

**Question 56 • Answer: E**

For the parallel pair,  $1/R = 1/6 + 1/3 = 1/6 + 2/6 = 3/6$ , so  $R = 2 \Omega$ . The total current is  $I = V/R = 12/2 = 6$  A; equivalently the branches draw  $12/6 = 2$  A and  $12/3 = 4$  A, summing to 6 A. The value 2 A accounts for only the  $6 \Omega$  branch and ignores the second resistor.

**Question 57 • Answer: D**

Energy = power  $\times$  time =  $100 \text{ W} \times 10 \text{ s} = 1000 \text{ J}$ . Once the bulb's power rating on that supply is given, the 200 V figure adds nothing; substituting the voltage number for the power would produce the spurious 2000 J.

**Question 58 • Answer: E**

Momentum is a vector, so the rebound reverses the sign:  $\Delta p = m(v_f - v_i) = 2 \text{ kg} \times (-3 - 5) \text{ m/s} = -16 \text{ kg}\cdot\text{m/s}$ , a magnitude of  $16 \text{ kg}\cdot\text{m/s}$ . The value  $4 \text{ kg}\cdot\text{m/s}$  comes from subtracting the speeds as plain numbers ( $5 - 3 = 2$ , times 2 kg) and forgetting the reversal of direction, the standard trap here.

**Question 59 • Answer: D**

Gravitation obeys an inverse-square law:  $g \propto 1/r^2$ . Going from the surface ( $r = R$ ) out to  $3R$  scales  $g$  by  $(1/3)^2 = 1/9$ , so the surface value is 9 times the orbital value:  $g = 2 \times 9 = 18 \text{ m/s}^2$ . The value  $6 \text{ m/s}^2$  comes from treating  $g$  as proportional to  $1/r$  instead of  $1/r^2$ .

**Question 60 • Answer: B**

$Q = mc\Delta T = 0.5 \text{ kg} \times 450 \text{ J/(kg}\cdot^\circ\text{C)} \times (100 - 20)^\circ\text{C} = 0.5 \times 450 \times 80 = 18,000 \text{ J}$ . The value 22,500 J arises from using  $\Delta T = 100^\circ\text{C}$  and ignoring the  $20^\circ\text{C}$  starting temperature.